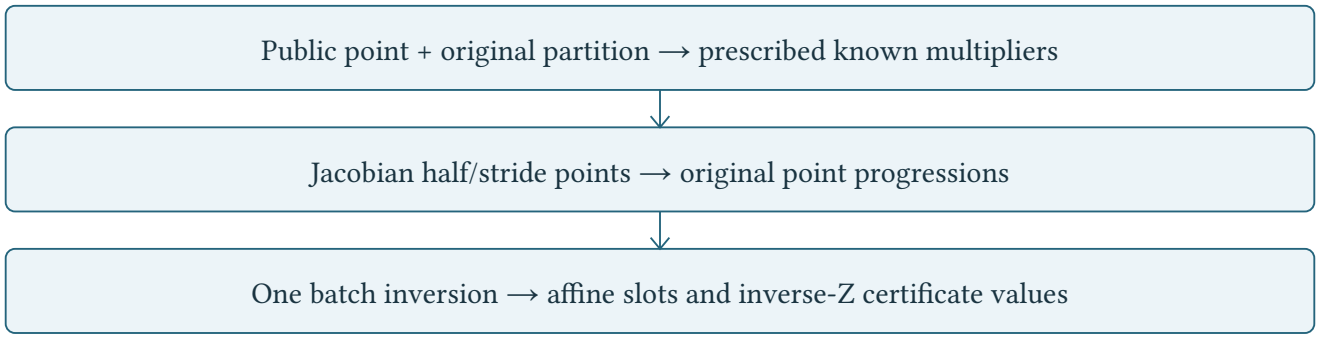
Native prescribed elliptic source grid

2026-11-03 · Exact public-generator checks · Individual-grid component

The native grid preserves the original public multiplier layout, Jacobian addition/doubling and one-inversion affine conversion. Its inputs are the public curve and point plus the source partition. It performs no unknown-scalar search.



71 saved slots matched | Next: mixed-stride shared window

Construction. Starting at the half-order multiple and a public stride, the code builds the original two progressions and optional tail point. Identity, vertical-pair, doubling and equivalent-scaled-coordinate cases are handled explicitly. The original partition and public point are validated.

Checked evidence. At source6c5f5b983d62, one standalone rustc1.99.1 build and one invocation passed all3 test methods. All71 slots across the three original grids match saved labels, Jacobian coordinates, affine coordinates and inverse-Z values exactly. Independent affine known-multiplier calculations agree. Additional checks cover exceptional points, rescaling, singleton conversion and rejection.

Conversion cost. For N nonempty finite points, prefix and reverse products plus coordinate conversion use7N-3 field products and one inversion, excluding validation/group construction. This preserves the original batch method; it is an algebraic count, not a measured speed improvement. Allocation and native integer costs remain. Empty/infinite batches are rejected.

Scope. The backend supports p up to half the u64 maximum; caller supplies primality and subgroup order. Fixed checks do not exhaust all points or field sizes. No fresh sign oracle, recovery cohort, cloud job or benchmark ran.

Next. Expose projective-only output and mixed-stride shared-window updates, then connect division/seed recurrences and the resultant stage. Repeating individual conversions must not replace the original shared-window accounting. The complete4,563-case cohort remains unchanged and unlaunched.

Evidence. Source and input locks, fixtures.tsv, slots.tsv, native build/test receipt and logs, and the byte-preserved original geometry source. See DERIVATION.md for exact formulas, costs, test limits and next integration.

Reference. EFD Jacobian formulas and original shared-projective-grid/geometry.py. Established formulas are reused; no fastest-formula or novelty claim is made.